Fiscal Year:	FY 2009	**Task Last Updated:**	FY 06/19/2009
PI Name:	Bruno, John Ph.D.		
Project Title:	Handheld FRET-Aptamer Sensor for Bone Markers		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Address 2:			
PI Web Page:			
City:	San Antonio	**State:**	TX
Zip Code:	78229-4253	**Congressional District:**	20
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	02/26/2009	**End Date:**	02/25/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Meck, J@n	**Contact Phone:**	281-244-5405
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX08CC16P		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Astronauts lose significant bone mass during lengthy spaceflights. Although, no effective treatments or prophylactics have yet been defined, it is important to monitor bone loss during missions. As such, the sensor must be compact and facile to operate in spacecraft. Operational Technologies Corp. (OpTech) proposes to complete development of its successful Phase I competitive fluorescence resonance energy transfer (FRET)-aptamer assays for several bone loss markers and calcidiol. In Phase I, OpTech developed several rapid (15 minutes) specific competitive polyclonal FRET-aptamer assays with low nanogram/ml sensitivity. OpTech also cloned and sequenced 110 bone marker and calcidiol aptamers which will be individually screened in Phase II for optimal FRET assay performance. During Phase II, OpTech will shift its FRET to emit in the red (> 600 nm) to avoid the intrinsic fluorescence of urine and serum. OpTech will also lyophilize and package optimized FRET-aptamer assays for use with a bubble-free plastic cuvette and body fluid collection system. The assay system will be coupled to a modified version of the commercial off-the-shelf (COTS) Picofluor[TM] handheld battery-operated fluorometer customized to detect bone markers and calcidiol in body		

	fluids. OpTech will deliver the packaged assays, handheld reader and software to NASA for testing. POTENTIAL NASA COMMERCIAL APPLICATIONS: NASA would use this technology as a portable means for astronauts to sensitively self-monitor their bone loss or bone preservation treatments with one-touch ease from urine or serum samples. Monitoring of other clinical analytes such as glucose and environmental microbe monitoring in water supplies by astronauts would also be possible, if appropriate FRET-assays were developed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Women and the elderly could use this system as a rapid and facile point-of-care (POC) diagnostic system to monitor osteoporosis and the efficacy of therapeutic regimens. The system might also be used to monitor bone repair following severe fractures or skeletal procedures related to plastic or reconstructive surgeries. In a broader sense, OpTech is developing FRET-aptamer assays for a wide array of analytes on Earth including foodborne pathogens and other bacterial, viral and clinical analytes.
Task Progress:	New project for FY2009. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)